

ENGLISH **LIVE**

45

COMPREHENSION & VOCABULARY

UPSC / BANK PO / SSC / OTHERS

SESSION - 45

30th MARCH, 2017



Battery Storage for Renewable Power

A burst of entrepreneurial energy down under promises to transform the fortunes of renewable power. Path-breaking entrepreneur Elon (Tesla) Musk is reportedly setting up a 100-MWhour power battery storage capacity in South Australia, in 100 days flat. It would be the world's largest utility-scale power storage — at an attractive cost — and slated to greatly boost renewable energy.

Also notable is the fact that Musk is leveraging scale economies to drive down the costs of electric power storage. The ballpark figure stated is \$250 per kWh as capital expenditure, and about \$0.3/kWh as operational-ex. Reports say that the Australia facility would use conventional lithium-ion batteries,



for which Tesla is joining hands with Panasonic. But costs should fall much further. There's a paradigm shift in the offering in battery storage technology. Vanadium-flow batteries promise to quite dramatically reduce the op-ex for electric storage to only about \$0.05/kWh, which would amount to grid-parity tariffs here,

in rupees. Note that the sheer intermittency of renewable energy, solar or wind power, makes battery storage crucial, with op-ex currently put at upwards of 30 cents/kWh. But this essentially ends up tripling the effective cost of renewable power generation, not counting evacuation costs.

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←
**DISCUSSION ON
30th Mar (Today)**

→
**DISCUSSION ON
31st March (tomorrow)**

Good idea, bad plan

DBT for fertilisers has to be similar to that on LPG

The fact that, as the newspaper reported on Wednesday, the government has been able to start various pilot projects on direct benefits transfer (DBT)—it has been done in 11 districts already—is a big positive and suggests that, over the next 6-12 months, it should be possible to cover the entire country; with 77% of point-of-sale outlets already equipped with Aadhaar-readingscanners, it may take even less time. The problem is that the 'DBT for subsidised fertiliser' project is loosely conceived and is unlikely to achieve its objectives, of reducing leakages of subsidies, encouraging the industry to grow and empowering farmers. What makes this a bigger pity is that the government already has a well-functioning DBT scheme for cooking gas (LPG); so, all that needed to be done was to copy that.

In the LPG scheme, customers get a fixed amount of subsidy in their bank accounts, and they pay the market price to the LPG supplier. Since all purchases are done at market prices, there are no dues for suppliers and this encourages new players to enter the market as well. In the case of fertilisers, however, purchases are made—based on an Aadhaar authentication of the buyer—at a price which is, right now, 30-40% of the market price; the balance is then credited to the suppliers' account. If this happens as soon as the sale happens, there is no difference—at the beginning of the year, dues to fertiliser firms exceeded ₹43,000 crore. With their fate in the hands of government policy on dues, it is unlikely too many new players would want to enter the market.

Also, in the case of LPG, there is a fixed number of cylinders everyone gets in a year, the subsidy is also known. For fertilisers, all that the DBT does is to authenticate who is buying it, there is no cap on the amount or the subsidy. Since 10-15% of fertiliser gets diverted to other countries or to industrial uses, in a DBT scenario, a farmer can buy 10 kg or 100 kg or 1,000 kg and be part of the diversion—ideally, then, the government must come up with per acre norms of usage and fix a ceiling on how much will be eligible for a subsidy; the subsidy per kg should also be fixed in advance. One argument against this is that, since all urea is neem-coated, the scope for diversion is limited—experts, however, argue that this may prevent urea from being mixed in milk, it does not stop it from being used in plywood manufacturing; to the extent, neem-coated urea makes it better, it reinforces the case for smuggling for use in farms in neighbouring Nepal and Bangladesh. Doing fertiliser subsidies the LPG way has another advantage in that it empowers farmers—China has moved to giving all input subsidies to farmers in cash. Once farmers get the subsidy in their bank accounts, they can decide how much of fertiliser they want to use, and even which one; but if the current system continues, farmers have no incentive to lower usage—this is why, right now, there is so much overuse of urea which also damages the soil. Indeed, the government also needs to move to LPG-style DBT for ration shops so the corrupt and distorting FCI system can be ended and farmers freed to grow the crop the market wants, not just what the government is able to buy.

FE The Financial Express

Date : 25th March, 2017

THE ECONOMIC TIMES

Date : 29th March, 2017



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Battery Storage for Renewable Power

THE ECONOMIC TIMES

Date : 29th March, 2017



Battery Storage for Renewable Power

- A **burst** of **entrepreneurial** energy down under promises to **transform** the fortunes of renewable power.
- **Path-breaking** entrepreneur Elon (Tesla) Musk is reportedly setting up a 100-MW hour power battery storage capacity in South Australia, in 100 days flat. It would be the world's largest utility-scale power storage — at an attractive cost — and **slated** to greatly **boost** renewable energy.
- Also notable is the fact that Musk is **leveraging** scale economies to **drive down** the costs of electric power storage. The **ballpark** figure stated is \$250 per kWh as capital expenditure, and about \$0.3/kWh as operational-ex.
- Reports says that the Australia facility would use **conventional lithium-ion batteries**, for which Tesla is joining hands with Panasonic. But costs should fall much further. There's a **paradigm** shift **in the offing** in battery storage technology.
- **Vanadium-flow batteries** promise to quite **dramatically** reduce the op-ex for electric storage to only about \$0.05/kWh, which would amount to grid-parity tariffs here, in rupees. Note that the **sheer** intermittency of renewable energy, solar or wind power, makes battery storage **crucial**, with op-ex currently put at upwards of 30 cents/kWh. But this **essentially** ends up tripling the effective cost of renewable power generation, not counting evacuation costs.
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COP21 • CMP11
PARIS 2015
UN CLIMATE CHANGE CONFERENCE

INDIA'S NDCs



LIVE

- 1 India aims to cut its carbon emissions **per unit of GDP by 33-35% by 2030 from 2005 levels**
- 2 Will achieve **40% of its cumulative installed capacity** of power generation from non-fossil based energy sources
- 3 Will create **carbon sink of 2.5-3 billion tonnes of CO₂** equivalent through additional forest and tree cover by 2030
- 4 **At least \$ 2.5 trillion** (at 2014-15 prices) will be needed to meet the goals by 2030



FOSSIL FUEL BASED POWER PLANT



Renewable Power





Battery Storage for Renewable Power

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A *burst* of *entrepreneurial* energy down under promises to *transform* the fortunes of renewable power.



Burst

A *burst* of start-up ecosystem is booming across the country.

**Burst**

A strong wish to do something

Burst



Entrepreneur

*Elon Musk is the serial **entrepreneur** who is into reusable rockets, electric vehicles as well as high speed transportation technology, Hyperloop.*



Entrepreneur

Someone who starts their own business to make money



Transformative

*Pradhan Mantri Gram Sadak Yojana is potentially a **transformative** project of rural India.*



Transformative

Causing a major change to someone or something to make it better



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Path-breaking

*The **path-breaking** research on reusable rockets is going to reduce the cost of launching satellites substantially.*



Path-breaking

Pioneering/innovative.





Slated

*The visit of
Bangladesh Prime
Minister to India is
slated in April 2017.*



Slated

Scheduled /Planned



Boost

Liberalisation
has given
boost to the
Indian
economy in
many ways.



Kerala State
Government
took several
measures to
boost its
tourism.



Boost

Increase / improve

Boost

Encourage / Help



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Leverage

*The organisation can reach greater heights, if it makes **leverage** of its resources.*



Leverage

To use an available resource, in order to achieve something better



Drive down

*The research and development in solar modules is **driving down** the cost of solar energy, but the problem is with regard to storage.*



Drive down

To cause something to decrease rapidly



Ballpark

*The **ballpark** figure is around \$ 16 bn for Mumbai-Ahmedabad bullet train project.*



Ballpark

Approximate cost or price



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Conventional

*Power generation through thermal power plants is an example of a **conventional** mode of power generation.*

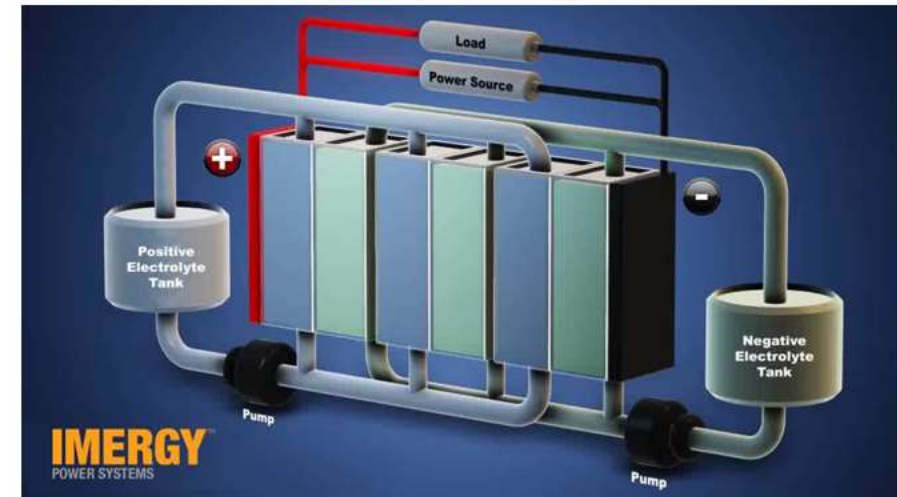


Conventional

Traditional / simple and ordinary / accepted or traditional ways of doing something



Lithium-ion batteries



Vanadium-flow batteries



Paradigm



Paradigm

A situation in which the usual and accepted way of doing things changes completely



In the offing

*World over, a big shift
towards electric vehicles is
in the offing.*



In the offing

Happening at sometime in the future / possible or likely to happen soon

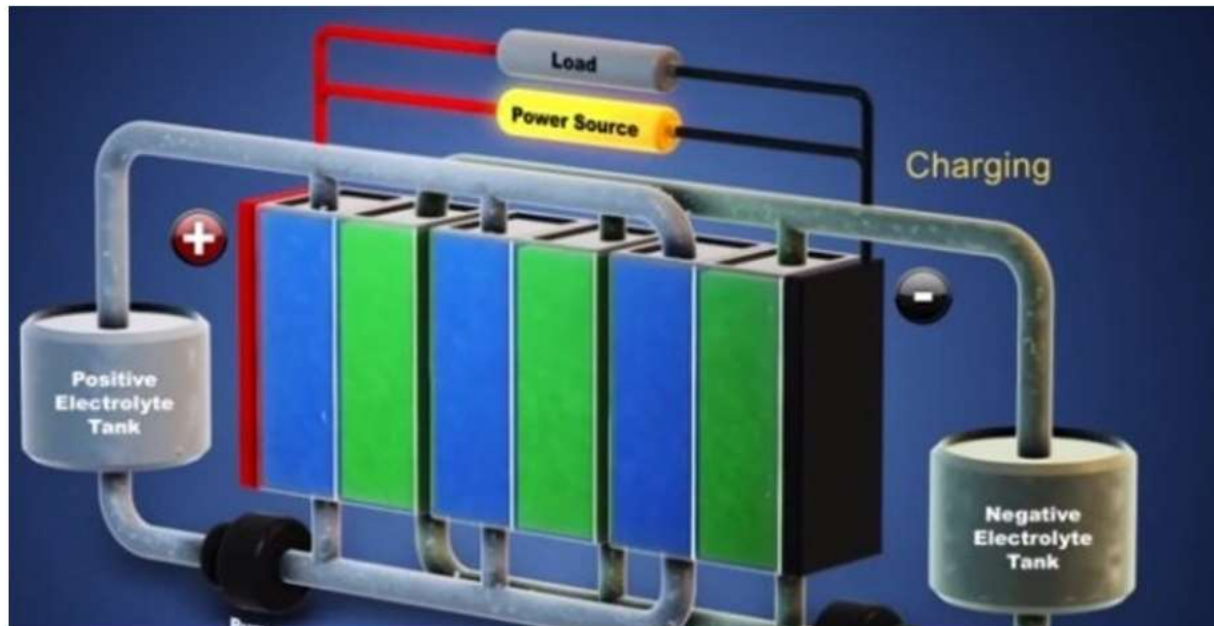


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Vanadium-flow batteries



Vanadium-flow batteries storage



Dramatic

*Tamil Nadu Politics took a **dramatic** turn as Sasikala Natarajan was convicted in disproportionate assets case.*



Dramatic

Sudden /unexpected thing to happen or full of action and excitement



Electric grid





Sheer



*The waterfall has a **sheer** drop of 50 meters.*



*His success in the UPSC examination was due to **sheer** determination.*

Sheer

Extremely steep

Sheer

Emphasis on greatness or importance



Crucial

Land acquisition is the **crucial** factor in the success of bullet train project.



Crucial

Extremely important or necessary



Essentially

*For penetration of digital transactions, seamless internet connectivity is **essential***



Essentially

Most important characteristics or ideas of something



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Non-flammable

*Biodiesel is sulfur-free,
non-flammable and
biodegrades faster than
sugar.*



Non-flammable

Not catching fire easily



Eminent

Chetan Bhagat is one among the **eminent** writers in India



Eminent

Famous, respected or important



Scalable

*Tech related companies
are normally scalable*



Scalable

A business, which can grow successfully larger with same methodology



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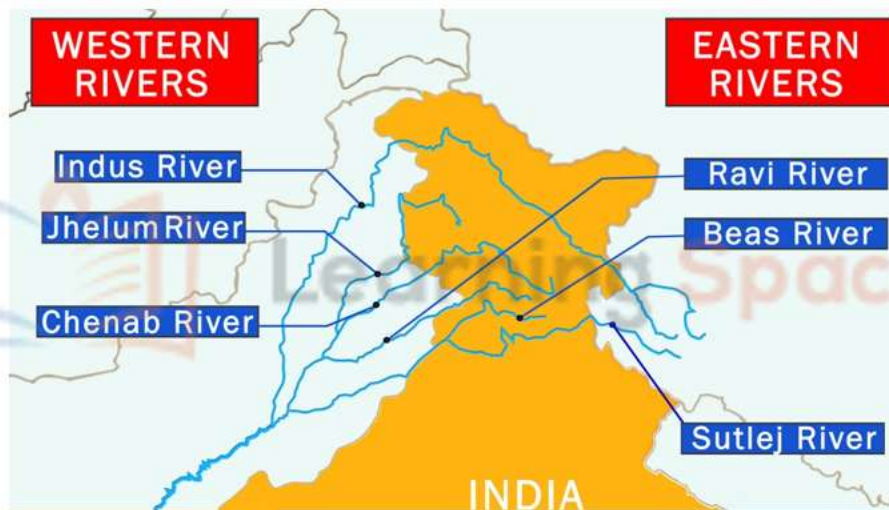
Proactive

Government officers must be more proactive, so as to deliver various schemes across the rural hinterland.



Proactive

Taking action to make changes yourself



*Indus Water Treaty is the **bilateral** water treaty between India and Pakistan.*

Bilateral



ASIAN INFRASTRUCTURE
INVESTMENT BANK

*These are **multilateral** financial institutions.*

Multilateral



Rev up

*Reliance Jio is getting
Revved up for the launch
of new products.*



Reliance
jio

Rev up

To become more active / to make someone or something become more active



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DISCUSSION ON 31st Mar (tomorrow)



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